WORKED SOLUTION

STEP I 2002, Question 10

Mechanics • Elastic Collisions

STEP I 2002, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q10

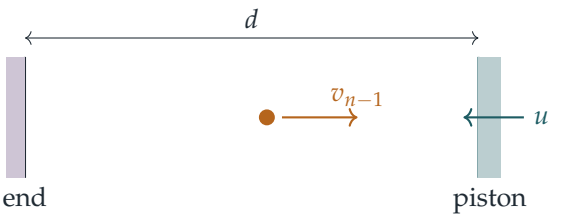
This question has a full guided treatment in *Beyond STEP*, Section 10.4, *A Molecule and a Piston*: https://beyondstep.keithmcnulty.org/ch10/piston.html

The key idea

A perfectly elastic collision with a wall that is far heavier than the particle reverses the *relative* velocity, so it is easiest to think in the frame of the piston: there the particle simply bounces off a stationary wall. Each bounce off the piston therefore adds $2u$ to the speed, while bounces off the fixed end change nothing. For the distances, notice that between two collisions with the piston the particle and the piston together cover a fixed pattern of lengths, which gives one linear equation for d_{n+1} in terms of d_n.

Solution

We model the air as a particle, the piston as a wall moving at constant speed u that is unaffected by collisions, and the end of the cylinder as a fixed wall. All motion is along the axis, so we take the direction from the end of the cylinder towards the piston as positive.



The speeds

Suppose the particle approaches the piston with speed w. The piston has velocity $-u$, so the particle's velocity relative to the piston is $w+u$ (towards it). In a perfectly elastic collision with a wall that is not deflected, Newton's experimental law with $e=1$ says the relative velocity is reversed, so afterwards the particle's velocity relative to the piston is $-(w+u)$. Its actual velocity is then

$$-u-(w+u)=-(w+2u),$$

so it leaves the piston with speed $w+2u$.

At the fixed end of the cylinder the same law, with the wall at rest, just reverses the velocity: the speed is unchanged. So between consecutive collisions with the piston the particle's speed stays the same.

Before the first collision the particle is moving towards the piston with speed v, so $v_1=v+2u$. Just after the nth collision it has speed v_n; it bounces off the end without change of speed and meets the piston again at speed v_n, leaving with $v_{n+1}=v_n+2u$. By induction,

$$v_n=v+2nu,$$

as required.

The distances

Between the nth and $(n+1)$th collisions (a time t_n) the piston moves inwards at speed u, so

$$d_n-d_{n+1}=u\,t_n.$$

In the same time the particle, moving at the constant speed v_n, travels from the piston to the end (a distance d_n, since the end is d_n from where the piston was) and back to the piston (a distance d_{n+1}, where the piston now is). So

$$d_n+d_{n+1}=v_nt_n.$$

Ask yourself How do we get rid of t_n?

Divide one equation by the other: $t_n>0$, so

$$\frac{d_n+d_{n+1}}{d_n-d_{n+1}}=\frac{v_n}{u}.$$

Cross-multiplying, $ud_n+ud_{n+1}=v_nd_n-v_nd_{n+1}$, so

$$(v_n+u)\,d_{n+1}=(v_n-u)\,d_n.$$

With $v_n=v+2nu$, we have $v_n-u=v+(2n-1)u$ and $v_n+u=v+(2n+1)u$, so

$$d_{n+1}=\frac{v+(2n-1)u}{v+(2n+1)u}\,d_n,$$

which is the required result.

A formula for d_n

Applying the recurrence repeatedly,

$$d_n = d_1 \prod_{k=1}^{n-1} \frac{v+(2k-1)u}{v+(2k+1)u} = d_1 \cdot \frac{v+u}{v+3u} \cdot \frac{v+3u}{v+5u} \cdots \frac{v+(2n-3)u}{v+(2n-1)u}.$$

The product telescopes: each denominator cancels with the next numerator, leaving

$$d_n = \frac{v+u}{v+(2n-1)u}\, d_1.$$

This also holds for $n = 1$, where it reads $d_1 = d_1$.

The case $v = u$

Putting $v = u$ gives

$$d_n = \frac{2u}{2nu}\, d_1 = \frac{d_1}{n}.$$

Then, from $ut_n = d_n - d_{n+1}$,

$$ut_n = \frac{d_1}{n} - \frac{d_1}{n+1} = \frac{d_1}{n(n+1)},$$

as required.

Checks

- With $u = 0$ the piston is a fixed wall: the formulae give $v_n = v$ and $d_n = d_1$ for all n, as they should.
- Take $u = 1$, $v = 3$, $d_1 = 75$. The formula predicts $d_2 = 50$, $d_3 = 37.5$, $d_4 = 30$ and speeds $5, 7, 9, 11$. An exact step-by-step simulation of the bouncing (in rational arithmetic) gives precisely these, and the same agreement holds for 200 random choices of u, v and starting position.
- When $v = u$, $\sum ut_n = d_1 \sum \left(\frac{1}{n} - \frac{1}{n+1}\right) = d_1$: the piston travels exactly d_1 in total, reaching the end of the cylinder, which is consistent with $d_n = d_1/n \to 0$.

Common slips

- Using $d_n - d_{n+1}$ wrongly, for example taking the particle's round trip as $2d_n$: the piston has moved in the meantime, so the return leg is only d_{n+1}.
- Adding u rather than $2u$ per collision. The relative velocity is reversed, so the change in the particle's velocity is twice the approach speed added by the piston.

Beyond STEP

Each collision changes the particle's speed by a fixed amount, but the collisions get more frequent as the gap closes, and the product $v_n d_n$ stays roughly constant for large n (here $v_n d_n \approx (v+u)d_1$ when n is large). Such a slowly-varying conserved quantity is an *adiabatic invariant*, and averaging over many molecules it explains why gas warms up when compressed quickly: for slow compression in one dimension, $T \propto 1/L^2$. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for collisions and reference frames.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.